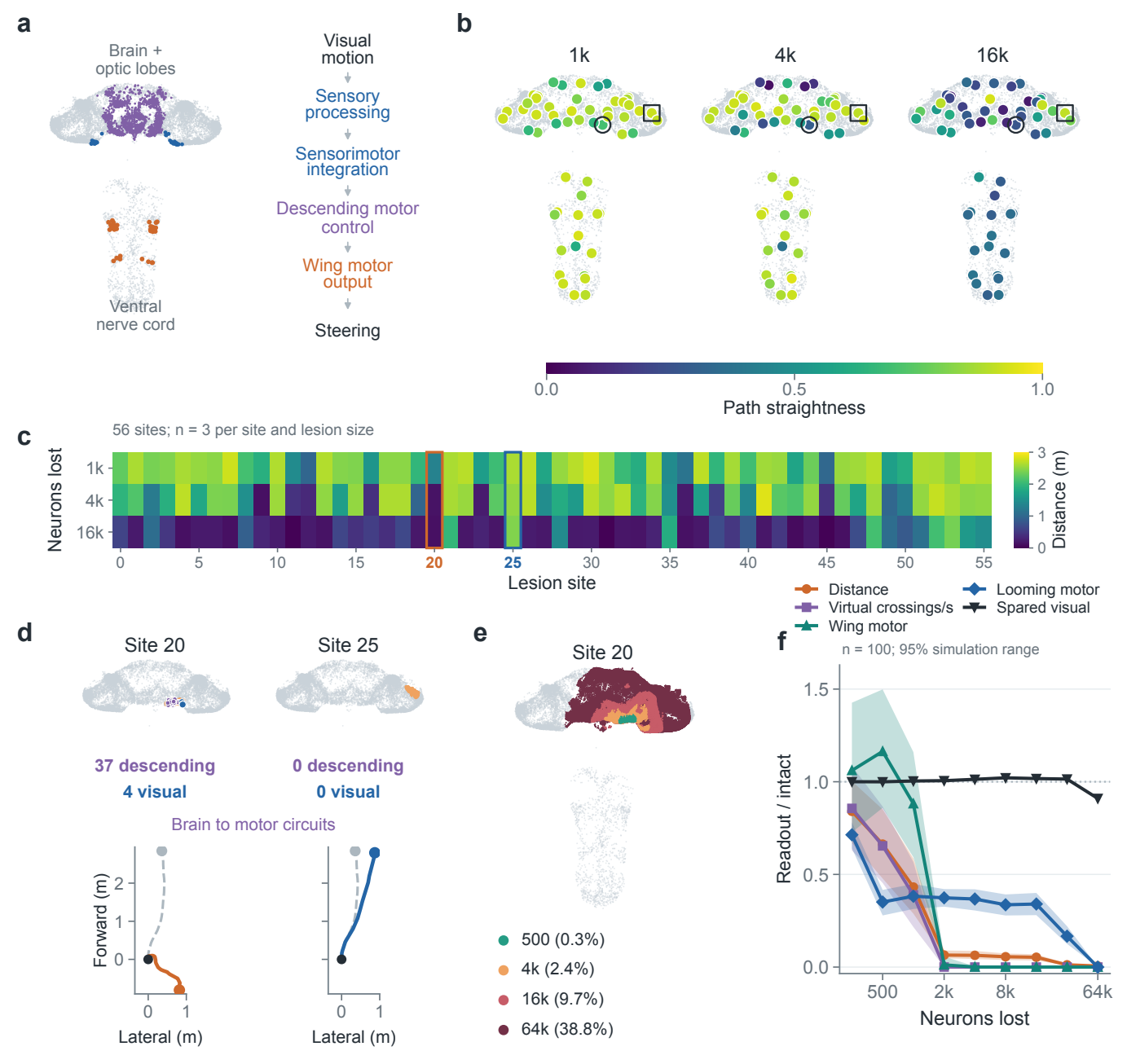

a
Brain + optic lobes
Ventral nerve cord
Visual motion
Sensory processing
Sensorimotor integration
Descending motor control
Wing motor output
Steering
b
1k
4k
16k
0.0
0.5
1.0
Path straightness
c
56 sites; n = 3 per site and lesion size
Neurons lost
1k
4k
16k
0
5
10
15
20
25
30
35
40
45
50
55
Lesion site
Distance (m)
3
2
1
0
Distance
Virtual crossings/s
Wing motor
Looming motor
Spared visual
d
Site 20
Site 25
37 descending
4 visual
0 descending
0 visual
Brain to motor circuits
Forward (m)
2
1
0
0
1
Lateral (m)
0
1
Lateral (m)
e
Site 20
500 (0.3%)
4k (2.4%)
16k (9.7%)
64k (38.8%)
f
n = 100; 95% simulation range
Readout / intact
1.5
1.0
0.5
0.0
500
2k
8k
64k
Neurons lost